

Fundamentals of Data Science

Lecture 4 - Data visualisation

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Some things about the exam

Take home exam

- 2 weeks: 4th - 18th December 2025, due at 12 noon UK time
- The exam will be a take home exam
- The exam jointly assesses D100 Fundamentals of Data Science and D400 Research Computing
- The exam will be similar to a data science challenge and you will be required to submit:
 - Your code
 - A report describing your modelling approach and presenting your results
- There will be a Q&A session (instead of a class) in week 9 on the 6th of December

Visualisations

Motivation

Motivation 1: Data Scientists visualise all the time

Motivation 2: Data vis. is not just descriptive statistics

We have four datasets and we can summarise them with the following descriptive statistics:

- $\text{mean} = 7.5$
- $\text{standard deviation} = 2$
- $\text{correlation coefficient} = 0.8$

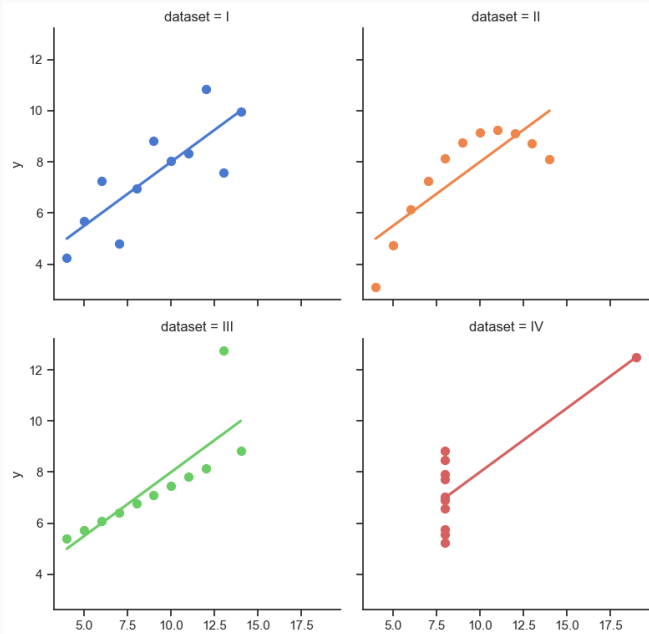
Motivation 2: Data vis. is not just descriptive statistics

We have four datasets and we can summarise them with the following descriptive statistics:

- $\text{mean} = 7.5$
- $\text{standard deviation} = 2$
- $\text{correlation coefficient} = 0.8$

→ So is it just the same data?

Anscombe's Quartet - data vis. is not just descriptive statistics

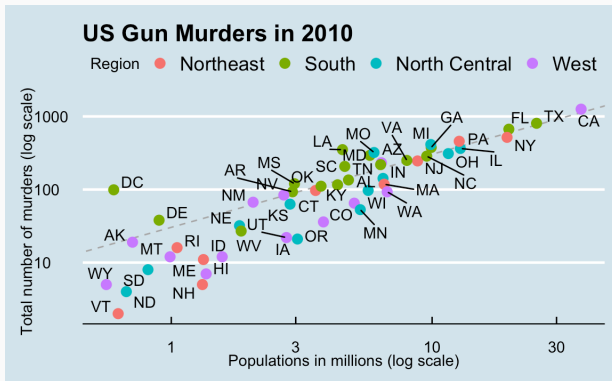


Motivation 3: Your data can tell you a lot

Gun murders in the US in 2010

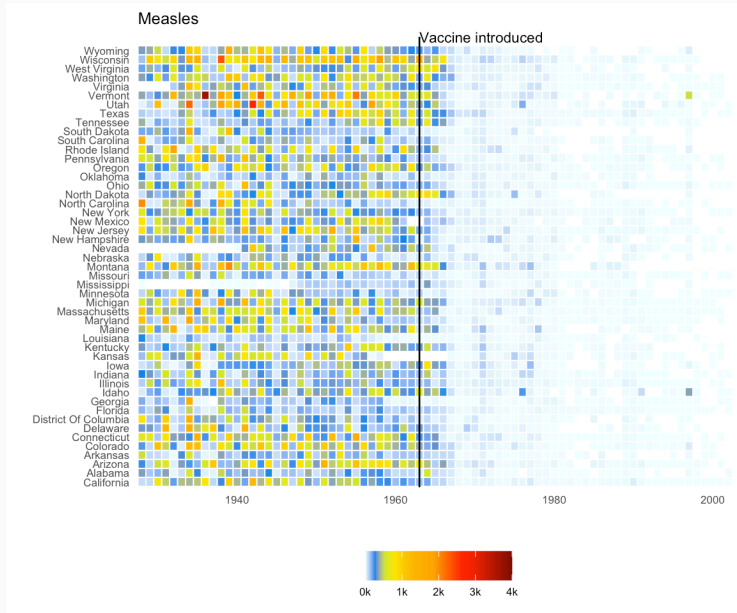
	state	abb	region	population	total
1	Alabama	AL	South	4779736	135
2	Alaska	AK	West	710231	19
3	Arizona	AZ	West	6392017	232
4	Arkansas	AR	South	2915918	93
5	California	CA	West	37253956	1257
6	Colorado	CO	West	5029196	65

...but even more when visualised



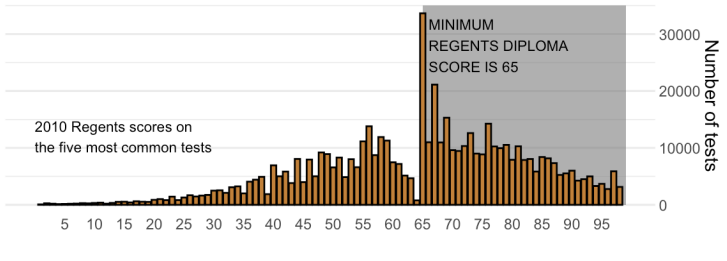
- The dashed gray line shows the relationship btw. population and murders
- States above (below) the line have higher (lower) murder rates than average for their population
- The log scales help display both small and large values clearly on the same graph

Motivation 4: Eye-balling statistics might be all you need

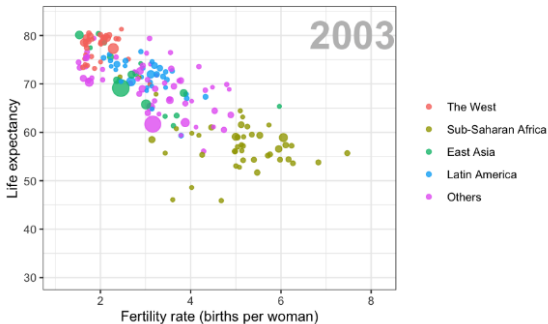


Motivation 5: Data vis. can lead to discoveries

Scraping by



Motivation 6: Communicate your results effectively



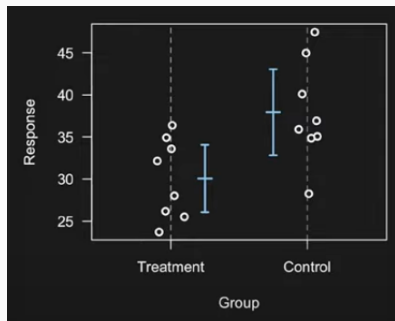
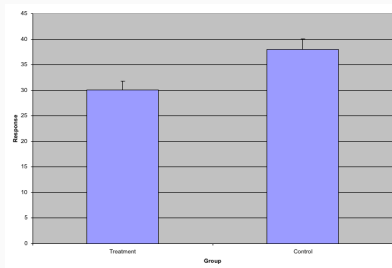
[Link to animated graph](#)

Best practices

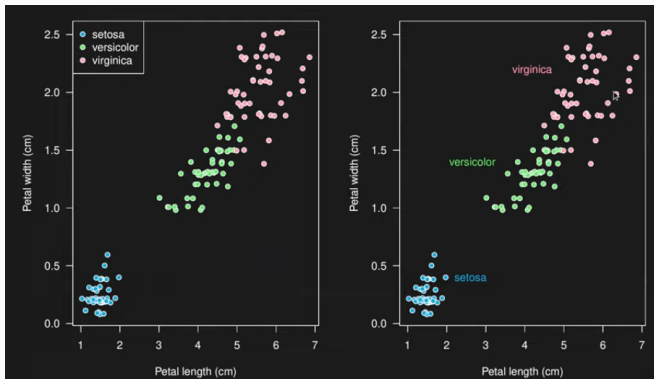
Best practice check-list

1. Show your data
2. Be accurate and clear
 - Label your axes with self-explanatory descriptions
 - Use a clear title that explains what is shown and the main message
 - Label your data directly and avoid legends
3. Sort by the variable in question
4. Reduce the data to the necessary
5. Use colour to set things apart (colour-blind-friendly)
6. Manage your proportions
7. Always include a zero for bar charts
8. Consider taking logs and/or differences
 - Generally, use linear scales for numbers and non-linear scales for rates of change
 - Log-scales can often help clean up lots of (positive) count data
9. Choose the right plot for your data and what you want to show
10. To compare plots:
 - Use common axis limits
 - Align things vertically
11. Keep your audience in mind

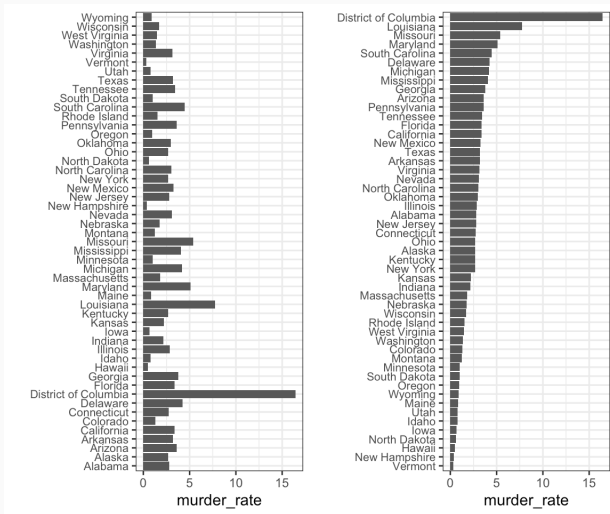
Show your data



Be accurate - label & avoid legends



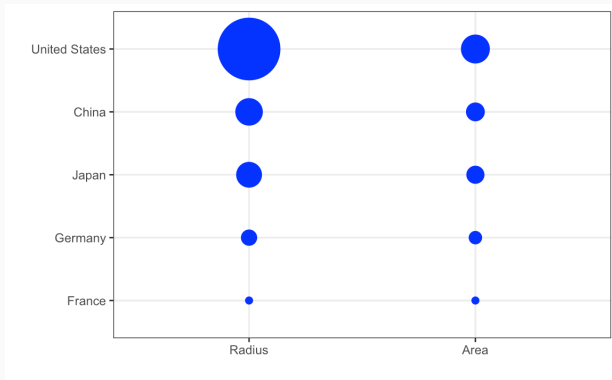
Don't sort by name, sort by what's relevant



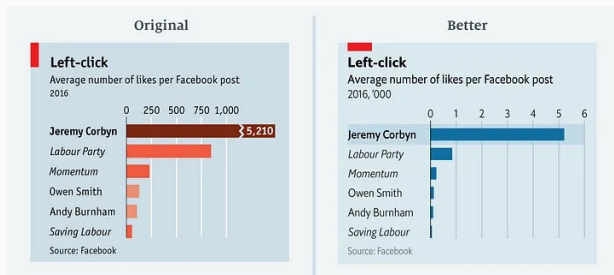
Manage your proportions



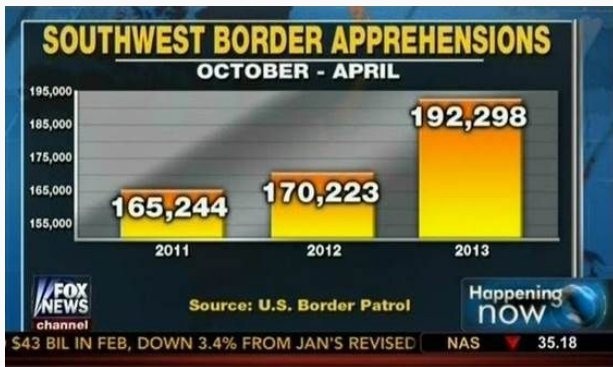
Proportions - use area and not the radius



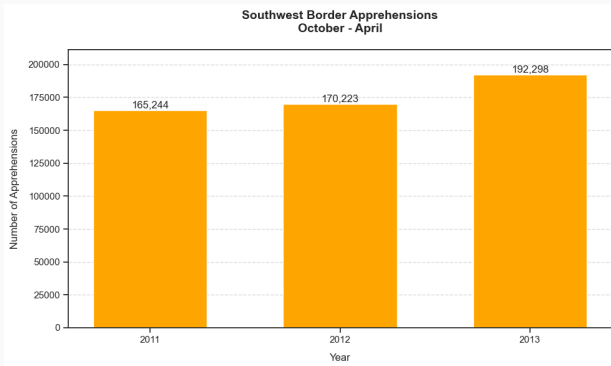
Manage your proportions



When to include zero? Always for bar charts!

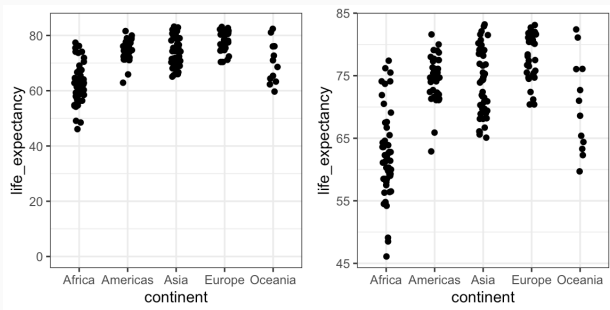


The difference a zero can make

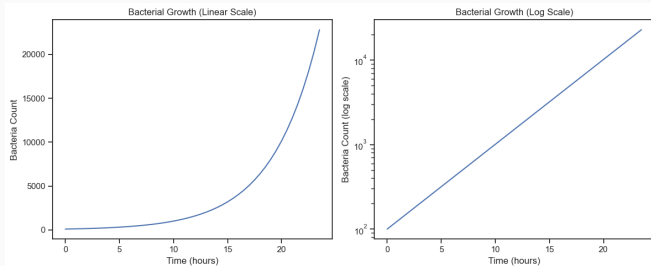


- Plot bar charts starting from zero.

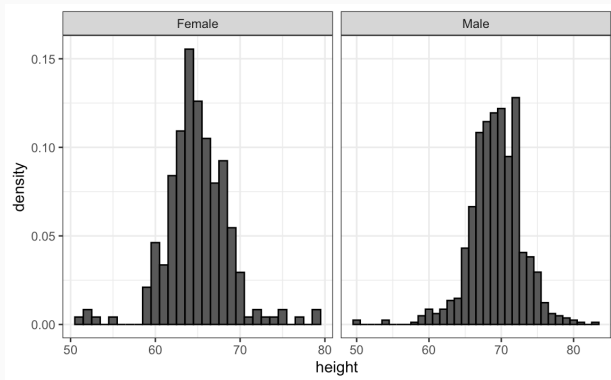
When to include zero? Maybe not for positional graphs



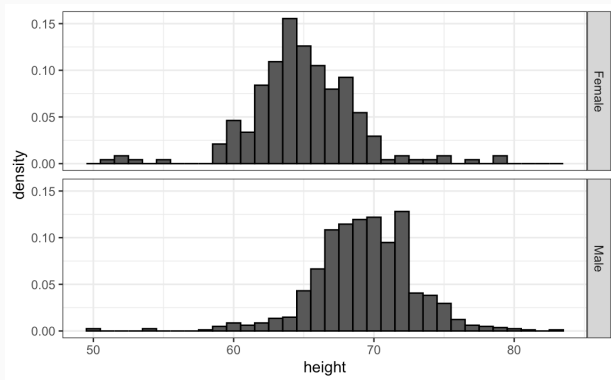
Use logs for growth



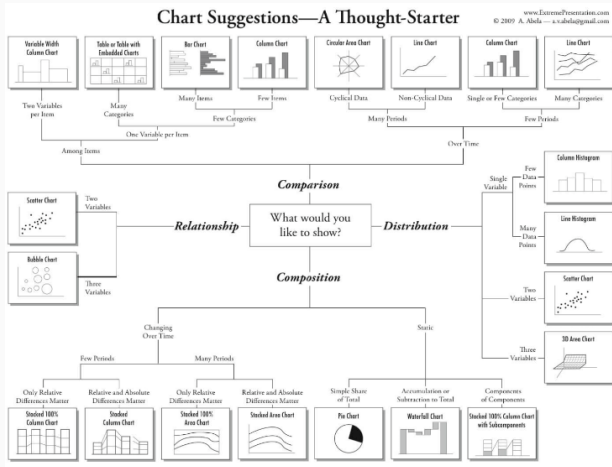
Compare graphs - the bad way



Compare graphs - same axes and vertically aligned



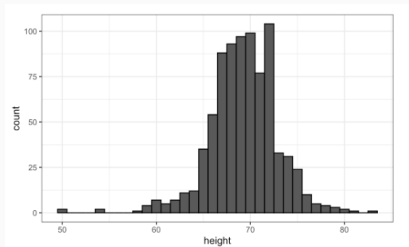
Choose the right graph



- Andrew Abela's chart suggestions
- FT graph choose
- Mostly bar, scatter, or line plots are enough

Some small tips

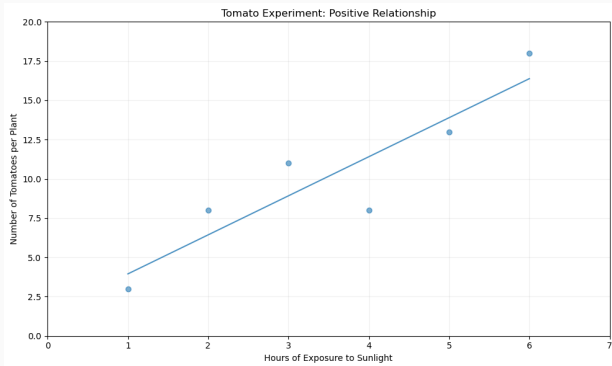
Use bar and count plots often



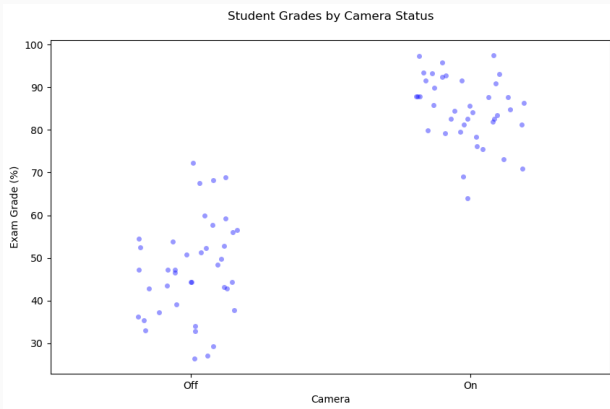
To get a quick feeling for your variable, use bar or count plots to check:

- range of values
- mean
- standard deviation
- outliers (do outliers drive results?)
- can I trust the data? are there values I did not expect?
 - e.g. someone is only 5 cm small or 8 meters tall.

Use scatter plots to get a feeling for correlations

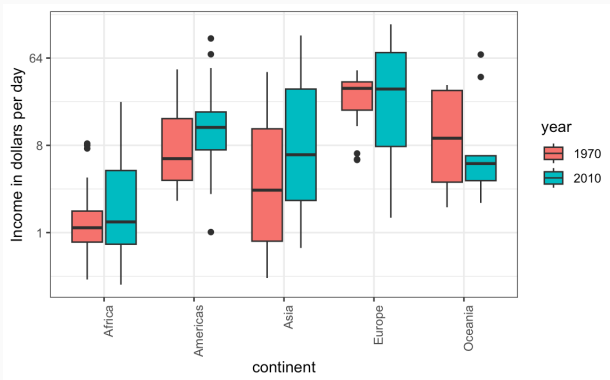


You can also do this for boolean variables



```
1 sns.stripplot(data=data, x='Camera', y='Grade', color='  
2 blue', alpha=0.4, jitter=0.2)
```

Compare changes for each group



Case study: look at your features and the prediction

target	price	product_id	age	insured
1	200	7	32	1
1	200	7	29	1

Case study: look at your features and the prediction

true	target	price	product_id	age	insured
0	1	200	7	32	1
1	1	200	7	29	1

Case study: look at your features and the prediction

true	target	price	product_id	age	insured
0	1	200	7	32	1
1	1	200	7	29	1

- Given your feature set, your model did not stand a chance to make the right prediction.
- Looking at these particular instances might reveal an important feature you forgot to include.

Case study: look at your features and the prediction

true	target	price	product_id	_product_id	age	insured
0	1	200	7	pregnancy product	32	1
1	1	200	7	pregnancy product	29	1

- Make it easy to interpret by mapping your columns back to descriptions.
- We then went to discuss our results with experts and for them the solution was obvious.

Case study: look at your features and the prediction

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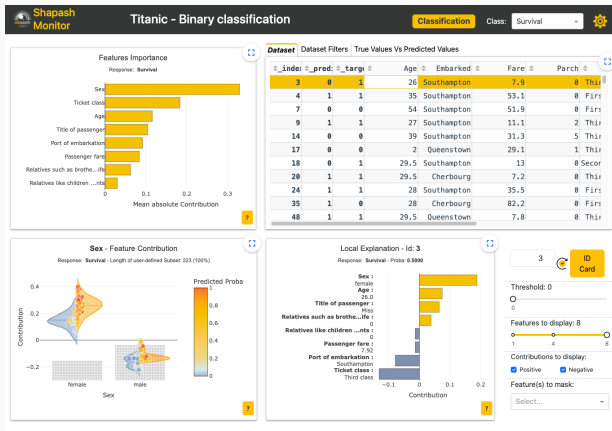
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- Add a gender column since the insurance would only provide this kind of product to females.

Case study: look at your features and the prediction

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0	1	200	7	pregnancy product	32	1
1	1	200	7	pregnancy product	29	1

- Make it easy to interpret by mapping your columns back to descriptions.
- We then went to discuss our results with experts and for them the solution was obvious.
- Add a gender column since the insurance would only provide this kind of product to females.
- NB: a fake example. You have to be careful in practice not to "discriminate" based on factors like age or gender.

Dashboards to interpret your model



Titanic dashboard

Plotting libraries

There are several plotting libraries in Python

- Matplotlib
 - Widely used and versatile plotting library.
- Seaborn
 - Built on top of matplotlib. Particularly good for statistical visualizations.
- Plotly
 - Used for interactive graphs and apps.
- Altair
 - Used for interactive graphs and apps.

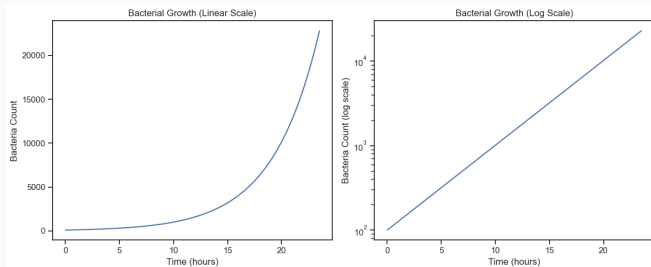
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→ Don't focus on one library but rather focus on the visualisation principles.

→ LLMs are great at helping you code the graph you want.

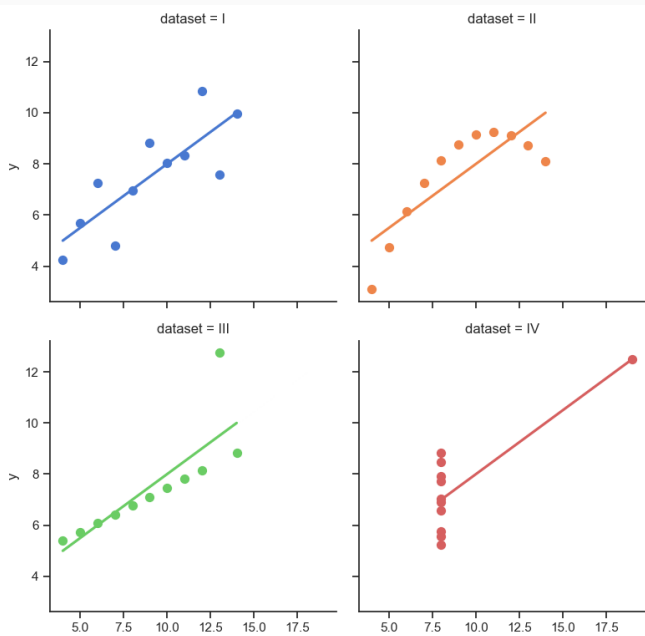
Remember our log example?



This was done in Matplotlib using synthetic (i.e. fake) data

```
1  import numpy as np
2  import matplotlib.pyplot as plt
3
4  # Generate data
5  time = np.arange(0, 24, 0.5)
6  bacteria_count = 100 * 2**(time/3) # Exponential growth model
7
8  # Create subplots
9  fig, (ax1, ax2) = plt.subplots(1, 2, figsize=(12, 5))
10
11 # Linear scale plot
12 ax1.plot(time, bacteria_count)
13 ax1.set_title('Bacterial Growth (Linear Scale)')
14 ax1.set_xlabel('Time (hours)')
15 ax1.set_ylabel('Bacteria Count')
16
17 # Log scale plot
18 ax2.plot(time, bacteria_count)
19 ax2.set_yscale('log')
20 ax2.set_title('Bacterial Growth (Log Scale)')
21 ax2.set_xlabel('Time (hours)')
22 ax2.set_ylabel('Bacteria Count (log scale)')
23
24 plt.tight_layout()
25 plt.show()
```

Remember Anscombe's quartet?

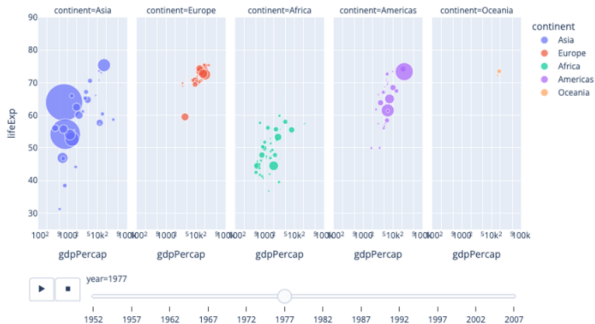


We used seaborn for this

```
1  import seaborn as sns
2
3  # Set the visual theme
4  sns.set_theme(style="ticks")
5
6  # Load the example dataset for Anscombe's quartet
7  df = sns.load_dataset("anscombe")
8
9  # Show the results of a linear regression within each dataset
10 sns.lmplot(
11     data=df, x="x", y="y", col="dataset", hue="dataset",
12     col_wrap=2, palette="muted", ci=None,
13     height=4, scatter_kws={"s": 50, "alpha": 1}
14 )
```

Plotly's interactive graphs

```
import plotly.express as px
df = px.data.gapminder()
fig = px.scatter(df, x="gdpPerCap", y="lifeExp", animation_frame="year", animation_size="pop", color="continent", hover_name="country", facet_col="continent", log_x=True, size_max=45, range_x=[100, 100000], range_y=[25, 90])
fig.show()
```



When and what to plot?

Explanatory vs. exploratory data analysis

Exploratory data analysis

- Private talk
- Less importance on best-practices
- Just make sure you really look at your data!

Explanatory vs. exploratory data analysis

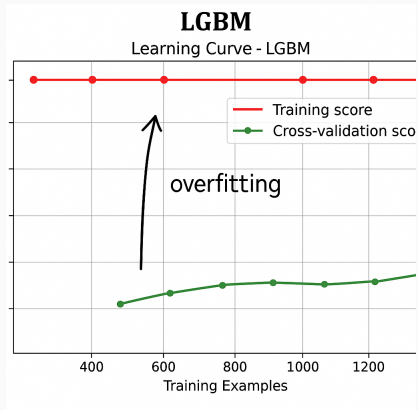
Exploratory data analysis

- Private talk
- Less importance on best-practices
- Just make sure you really look at your data!

Explanatory data analysis

- Who is my audience?
- Am I showing the whole picture?
- What do I want to achieve?
- Importance of best-practices

Please explain your graphs



- Every graph in the report should be explained.
- Make sure to explain anything that looks odd.

- Let's have another look at the titanic notebook.
- And remember all of those visualisations.

Titanic example

- Let's have another look at the titanic notebook.
- And remember all of those visualisations.

Titanic example

In the exam:

- Don't just throw a lot of those graphs at us.
- Do the exploratory analysis in your repo.
- But for the report, we are looking for the explanatory analysis where you generate insights and convince us of your modelling approach.

We should know why you chose those features

all of this and only those two predictors?

```
predictors = ["Sex", "Pclass"]  
target = 'Survived'
```

Define the training and validation data and labels.

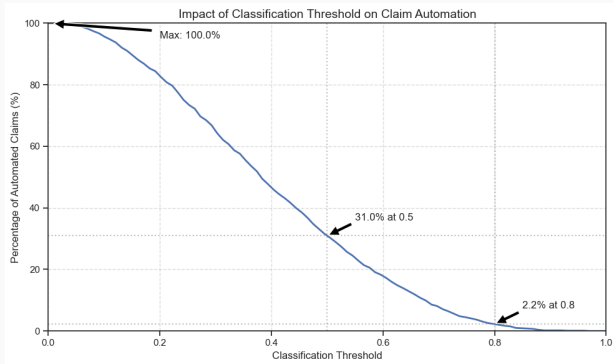
```
train_X = train[predictors]  
train_Y = train[target].values  
valid_X = valid[predictors]  
valid_Y = valid[target].values
```

Initialize the classification algorithm.

```
clf = RandomForestClassifier(n_jobs=-1,  
                           random_state=42,  
                           criterion="gini",  
                           n_estimators=100,  
                           verbose=False)
```

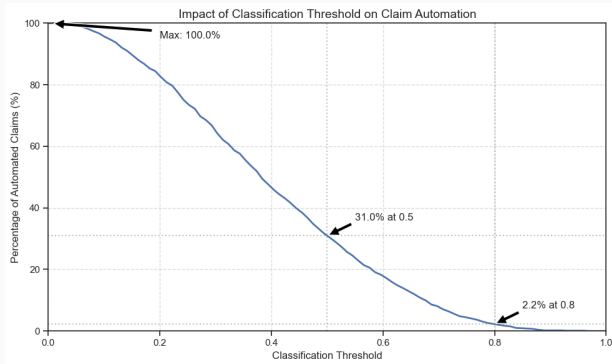
- For example: when we analyse your model's feature set, your **explanatory** data analysis should inform this feature selection.

Case study: showing the effect of your DS solution



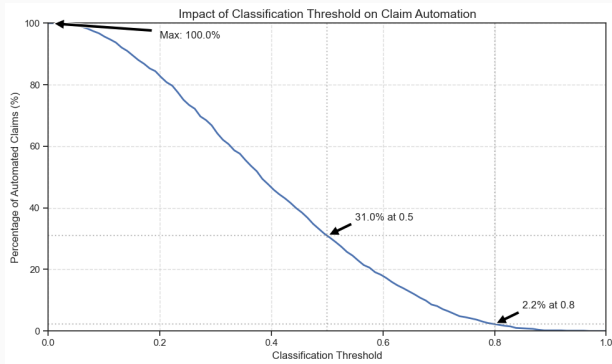
- How would you choose the threshold?

Case study: showing the effect of your DS solution



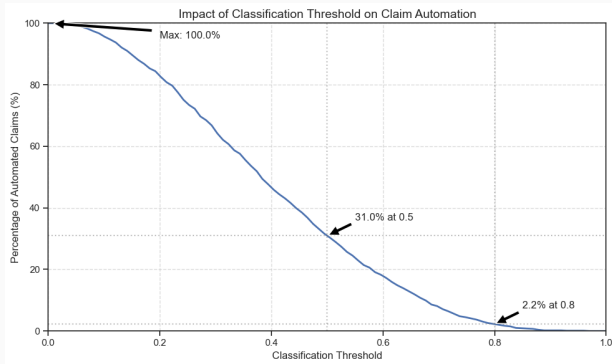
- How would you choose the threshold?
- It's not really possible!

Case study: showing the effect of your DS solution



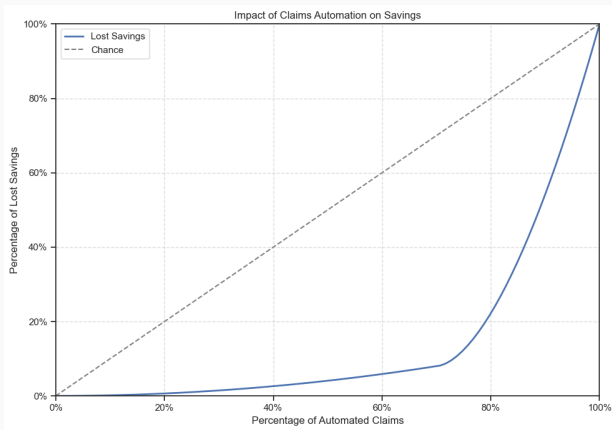
- How would you choose the threshold?
- It's not really possible!
- We did not build the right solution!

Case study: showing the effect of your DS solution

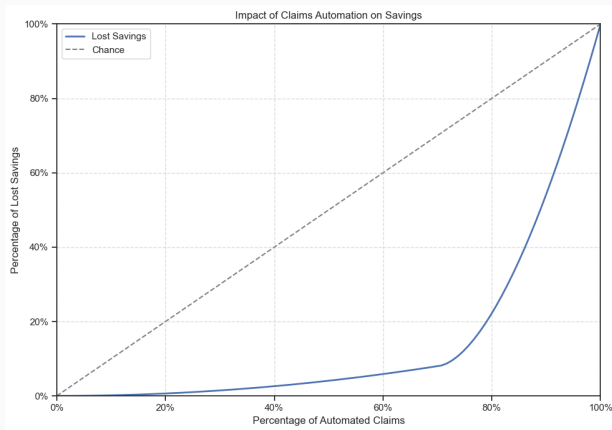


- How would you choose the threshold?
- It's not really possible!
- We did not build the right solution!
- We need to know how much money we'd lose when automating more claims.

Case study: a new model and a new graph!



Case study: a new model and a new graph!



- This allows us to take decisions!

Interpret your results

Table: Model Performance Comparison

Metric	GLM	LGBM
Mean Absolute Error (MAE)	183,404 AUD	146,942 AUD
Root Mean Squared Error (RMSE)	266,297 AUD	224,149 AUD
Mean Prediction	1,093,833 AUD	1,087,174 AUD
Predictive Accuracy (MAE ÷ True Mean)	16.9 %	13.6 %

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- Is 13% good?
- Is this a useful model for your use case?

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Predictive Accuracy (MAE ÷ True Mean)	16.9 %	13.6 %

- Is 13% good?
- Is this a useful model for your use case?
- Write something like: "While the LGBM achieves a relative mean absolute error of 13.6%, corresponding to an average deviation of about AUD 147k, such a margin may limit its suitability for production use in individual house price estimation. However, the model could still be valuable for ranking or segment-level valuation."

Some SWE stuff again

Your code should run out of the box

For the exam, you should be able to send us your repo and following your readme, we should easily get your code up and running by:

- Installing the environment
- Following data load instructions (should there be any) in Readme
- Executing your training pipeline
- Executing your Jupyter notebook

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That means:

- Clear instructions in Readme
- Environment yaml should exist and include all needed packages
- Imports of your own modules should work (pyproject.toml)
- Paths are setup in a "computer" agnostic way

Pathlib library

Do not use `"/User/adrian/data/nba.csv"`

We will not be able to load the data if you use paths like

```
1 df = pd.read_csv("/User/adrian/data/nba.csv")
```

```
2
```

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or this, depending on from where we execute the script.

```
1 df = pd.read_csv("data/nba.csv")
```

```
2
```

or if we work with a Mac or Windows:

```
1 Mac formatting: /Volumes/Share/Main Folder/Folder1/Folder2/  
FinalFolder
```

```
2 Windows formatting: S:\Main Folder\Folder1\Folder2\  
FinalFolder
```

```
3
```

Pathlib is a Python package (comes directly with Python) that helps you avoid these problems by:

- Easy path manipulation (parent directories, file name, suffix)
- Resolving relative paths to absolute paths
- Cross OS compatibility (Mac, Windows)

Moving to the root directory

```
├── README.md
├── config.yaml
├── data
│   └── game.csv
├── environmnet.yaml
├── nba
│   ├── __init__.py
│   ├── config.py
│   ├── data_load.py
│   └── training_models.py
└── pyproject.toml
```

Say we are in the "data_load.py" script and want to load data from the data folder:

```
1  from pathlib import Path
2  path = Path(__file__).parent.parent / "data" / "game.csv"
3  print(path)
4  # Output: /Users/adrianochs/repos/cambridge_ds/
5  config_example/data/game.csv
```

- We navigate from the script to the root and then to the ".csv" file
- NB: we get the absolute path

Graphs were borrowed from:

- R. A. Irizarry (2022) “Introduction to Data Science”, Chapter 7
- Talk by K.W. Broman (2017) “Creating effective figures and tables”
- Article by S. Leo (2019) from The Economist
- Article by Lars Roemheld (2023) on Towards Data Science